

A Systematic Review of Sensor–AI Integration in Structural Health Monitoring of Civil Buildings

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Abstract

Structural health monitoring (SHM) is a component of modern civil engineering. This review analyzes the integration of sensing technologies and artificial-intelligence-based methods for damage detection, localization, classification, prognosis, and anomaly detection in buildings and civil infrastructure. The database search covered Web of Science (WoS), Scopus, and IEEE Xplore for the period 1 January 2020–31 December 2025. The initial records were 292 in WoS, 311 in Scopus, and 338 in IEEE Xplore; after applying the AI-related search constraint, the corresponding AI-SHM corpora were 71, 79, and 139 records, respectively. The combined screening and eligibility workflow produced 31 open-access studies for detailed qualitative analysis, while the task-specific performance tables synthesize the subset of studies for which the sensor type, AI model, SHM task, validation context, and performance metrics were explicitly reported. The review, therefore, interprets reported performance by SHM task and sensor modality, rather than treating heterogeneous metrics as directly comparable across different datasets and experimental conditions. The results indicate that high values reported for accelerometer-, fiber-optic-, piezoelectric transducer-, and vision-based systems are mainly obtained under controlled, benchmark, simulated, or study-specific validation conditions. Consequently, robustness, transferability to operational structures, uncertainty quantification, sensor-network design, and integration with Physics-Informed Machine Learning and Digital Twin technologies remain central research needs.

Keywords: machine learning; deep learning; smart sensors; fiber bragg grating; optical fiber sensors; piezoelectric transducer sensors; accelerometers; damage detection; civil engineering



Academic Editors: Gaetano Miraglia and Stefania Coccimiglio

Received: 16 May 2026

Revised: 2 June 2026

Accepted: 5 June 2026

Published: 8 June 2026

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1. Introduction

Structural health monitoring (SHM) is the process of evaluating the condition of a structural system in real time or near real time. This analysis uses data collected from sensor sources to identify potential damage [1,2]. An SHM system consists of four components: sensor instrumentation for data collection, a data acquisition system for converting analog signals into digital format, advanced data analysis algorithms for extracting information, and reporting mechanisms for communicating the results to engineers and managers for making decisions regarding infrastructure maintenance and safety [2,3]. Sensors form the foundation of any SHM system, as they convert the physical quantities of the structure

into interpretable electrical or optical signals [2]. The monitored structures reported in the literature include bridges, tunnels, office buildings, civil constructions, and nuclear power plants, etc. These sensors can be classified based on their physical measurement principle, such as optical, piezoelectric, imaging-based, based on other electrical properties, or on acoustic properties.

The most commonly used SHM technologies employ optical fiber sensors (OFS), which have become ideal transducers due to their superior characteristics [4]. Their operating principle is based on the modulation of the light properties (frequency, amplitude, phase) of a light beam propagating through an optical fiber following interaction with external parameters of the structural environment, such as deformation or temperature [5]. OFSs are immune to electromagnetic interference (EMI), making them suitable for harsh industrial environments or near electrical equipment [5].

Fiber Bragg grating (FBG) sensors are a subcategory of OFS. It works by creating a periodic diffraction structure in the core of the optical fiber. This reflects a specific wavelength, called the “Bragg wavelength” [6]. Any change in strain or temperature at the sensor level leads to a change in the grating period and refractive index. Measuring this displacement allows for the calculation of the magnitude of the external disturbance. The accuracy of strain measurement is faithful to reality, ranging from ± 2 to ± 3 microstrain for commercial models [7]. The sensitivity to strain varies between 0.64 pm/microstrain and 1.2 pm/microstrain, and to temperature, between 6.8 pm/°C and 13 pm/°C [6]. In practice, a disadvantage of FBG is crosstalk between strain and temperature measurements, as both cause wavelength shifts. Common solutions for temperature compensation include using a non-strained reference sensor near the measurement sensor or designing specialized devices [8].

Other technologies use optical sensors and are identified as Raman-based Distributed Temperature Sensing (DTS), Rayleigh-based Distributed Acoustic Sensing (DAS), and Brillouin Optical Time-Domain Analysis (BOTDA), which is based on Brillouin scattering. These measure physical parameters along the entire length of the optical fiber, not just at discrete points. These technologies are ideal for monitoring long elements such as pipelines, tunnels, and dams because they provide continuous data on temperature, vibration, or strain [5]. Piezoelectric transducer (PZT) sensors are used for such measurements. They are capable of converting mechanical energy into electrical energy and vice versa [4,9,10]. In SHM applications, PZT generates and detects ultrasound waves, analyzing changes in the trajectory and shape of the reflected wave to detect micro-cracks or internal delamination [11]. They are known for their resilience in extreme conditions, such as intense cold, and are used in cryogenic monitoring systems [2,12].

Another area is image-based sensors, which includes computer vision and Light Detection and Ranging (LiDAR) [11,13]. They use standard cameras or laser technologies to capture images of the structure’s surface, and then image processing algorithms are used to identify and characterize visible anomalies, such as cracks [14,15]. LiDAR creates three-dimensional (3D) models of structures for monitoring deflections and inclinations [11]. The use of drones (unmanned aerial vehicles-UAVs) equipped with these sensors accesses hard-to-reach infrastructure [2].

Civil engineering includes buildings and infrastructure works intended for public or private use, excluding industrial constructions and those designed for specific tasks. Figure 1 presents the main categories of civil structures, along with the sensors used in SHM monitoring.

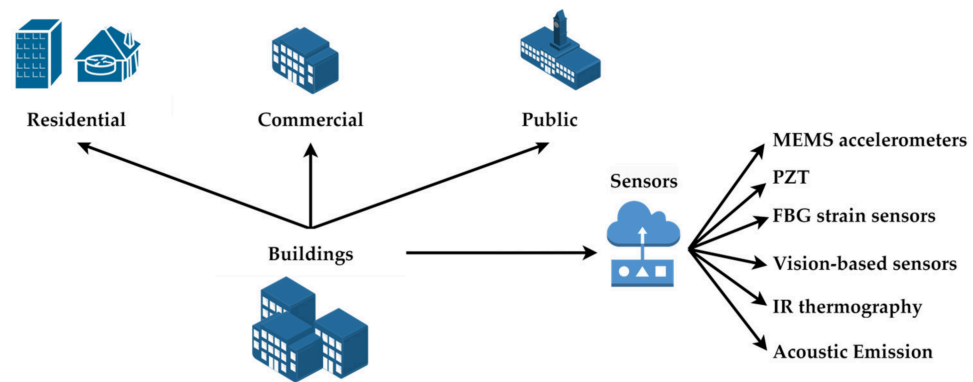


Figure 1. Classification of the main types of civil structures and the sensors used in SHM.

Residential buildings include houses, apartment buildings, and residential complexes. The public buildings category targets schools, universities, hospitals, administrative buildings, museums, and theaters. Commercial buildings are related to office buildings, shopping centers, and hotels.

Building SHM combines sensors targeting structural response, deformations, damage detection, and environmental influences. Structural response monitoring is performed using micro-electromechanical systems (MEMS) accelerometers. These are used to capture ambient vibrations and seismic response, while piezoelectric accelerometers are used for modal identification. Deformation and strain monitoring are carried out using FBG sensors. They measure deformations. Also in this category of measured parameters are electrical strain gages for evaluating local deformations and displacement sensors, such as linear variable differential transformers (LVDTs) or laser sensors. The latter monitors the drift between levels and settlements. Damage detection is performed by PZT sensors, which are used in SHM for crack detection. Alongside the PZT sensors are acoustic emission (AE) sensors, used to identify crack initiation and propagation. Environmental and thermal effects are monitored using temperature sensors. Their objective is to compensate for the thermal effects of infrared (IR) thermography sensors used to detect thermal anomalies, moisture infiltration, and delamination. Vision-based monitoring systems (video cameras and digital image correlation (DIC) techniques) are used for non-contact measurements of crack opening and surface deformations.

This article aims to conduct methodological research on the integration of sensors into SHM systems specific to buildings and civil infrastructure in service. The sensors included in the analysis are strain gages, accelerometers, OFS, FBG, and inclinometers. The research directly targets the use of artificial intelligence (AI), Machine Learning (ML), Deep Learning (DL), Neural Networks (NN), and anomaly detection tools. The methodology employs a combined approach, integrating quantitative and qualitative methods. The first investigation analyzed a statistical synthesis of indexed literature in Web of Science (WoS), Scopus, and Institute of Electrical and Electronics Engineers (IEEE) Xplore databases between 1 January 2020 and 31 December 2025. The second investigation focuses on a thematic analysis of the AI transformation related to sensor data processing. It will also include an explicit comparison between traditional structural monitoring methods and AI-augmented solutions. The comparison highlights the favoring of certain AI models in scientific investigations, their advantages, limitations, ethics, and contexts of applicability. The timeframe is limited to the period 2020–2025 to highlight the trends of researchers in the field, referring to modern AI technologies. Scientific sources refer to scientific articles and reviews from the WoS, Scopus, and IEEE Xplore databases.

Specialized literature usually addresses either the sensor technologies used in SHM or the applications of AI in structural data analysis. Therefore, there are a limited number of

papers that systematically analyze the interdependence between the type of sensors, the types of data generated, the appropriate AI architecture for each SHM task, the constructed datasets (public or private), as well as the performances achieved by models specific to the chosen architecture. In the category of SHM tasks, detection, localization, classification, and prediction are identified. The identified gap arises from the absence of a synthesis that correlates the hardware component, represented by sensors, with the software paradigm, represented by the AI model. Another gap concerns the methodological perspective in relation to the performance indicated in the specialized literature. Through this paper, the authors aim to address the interface between sensors and AI by concurrently analyzing these two dimensions and investigating how they co-evolve in SHM applications for civil infrastructure.

The specific contribution of this review is the explicit linkage between sensor modality, SHM task, AI model family, validation context, and reported metric. This framing differentiates the paper from surveys that discuss sensors or AI models separately, because it shows where the hardware-software pairing is mature, where evidence remains limited to controlled or benchmark conditions, and where deployment barriers remain insufficiently solved for operational civil infrastructure.

To demonstrate this novelty more explicitly, Table 1 positions the present review against previous SHM-AI review orientations. There are 7 criteria used for comparison: sensor-AI linkage, SHM task taxonomy, validation context, metrics compared, evidence grading, civil buildings, and open access PRISMA.

Table 1. Comparison of this review with representative existing SHM-AI surveys.

Reference	Sensor–AI Linkage	SHM Task Taxonomy	Validation Context	Metrics Compared	Evidence Grading	Civil Buildings	Open Access PRISMA
Alwis et al. [16]	No	No	No	No	No	Yes	Yes
Senkans et al. [17]	No	No	No	No	No	No	Yes
Kot et al. [18]	Partial	No	No	No	No	Yes	Yes
Tao et al. [19]	No	No	No	Partial	No	No	Partial
Vijayan et al. [20]	Partial	Partial	No	No	No	Yes	Yes
Rabi et al. [21]	Partial	Partial	Partial	Partial	No	Partial	Yes
Júnior et al. [22]	No	Partial	No	No	No	Yes	Yes
Cha et al. [23]	Partial	Yes	Partial	Partial	No	Partial	Partial
Markom et al. [24]	No	No	No	No	No	Partial	Partial
Sun et al. [25]	No	Partial	No	No	No	Partial	Yes
Bianchi et al. [26]	Partial	Partial	No	No	No	No	Partial
Kaveh and Alhajj, [27]	No	No	No	No	No	Yes	Partial
Nasir et al. [28]	No	Partial	No	No	No	Partial	Partial
Current review paper	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: Sensor–AI linkage shows the explicit pairing of the sensor modality with the AI model for each SHM task. SHM task taxonomy refers to a structured detection/localization/classification/prognosis framework. Validation context points to experimental conditions reported per study. The metrics compared indicate that a cross-study performance comparison was attempted. Evidence grading denotes a formal confidence- or evidence-level framework applied. Civil buildings means the review scope explicitly includes civil buildings (not only bridges/aerospace). The Open Access PRISMA corpus indicates that PRISMA-compliant screening is restricted to open-access full texts.

The paper aims to conduct a methodological analysis regarding the integration of sensors into SHM systems. These are applied to civil constructions, with the aim of studying AI technologies in this field. The paper highlights how hardware components have evolved through sensors in relation to software components through AI algorithms.

The objectives of the paper are as follows:

- Identifying the main types of sensors used in SHM systems for civil structures;
- Synthesizing the detection and data acquisition technologies used in SHM;

- Implementing a mixed methodology that combines a quantitative analysis of the scientific literature from the WoS, Scopus, and IEEE Xplore databases;
- Qualitative analysis of papers from the specialized literature focusing on AI applications in sensor data processing applied in SHM;
- Comparing traditional structural monitoring methods based on physical and analytical models with modern AI-augmented solutions;
- Identifying overstudied AI models, the performance achieved, current limitations, research gaps, and contexts of applicability for each approach are analyzed.

The paper contributions are listed below:

- Conceptualizing the theoretical framework;
- Defining the systematic literature review methodology;
- Developing the search strategy and selecting scientific articles from the WoS, Scopus, and IEEE Xplore databases;
- Conducting the quantitative bibliometric analysis;
- Qualitative analysis of the selected studies;
- Writing and structuring the article to highlight the interpretation of the results in order to provide conclusions that researchers can integrate into future studies.

In relation to the paper's objectives, the research questions are as follows:

- How does the type of sensor influence performance and the choice of AI architecture in different SHM tasks?
- What are the predominant patterns used in correlating sensor data with ML/DL methods reported in the recent literature?
- What are the technical and methodological limitations in integrating sensors with AI for monitoring civil infrastructure?

The article is structured into eight sections. Section 2 corresponds to the research methodology and includes the literature selection strategy along with the analysis criteria. Section 3 presents the results of the quantitative analysis, whereas Section 4 describes the classification of SHM applications. Sections 5 and 6 focus on qualitative analysis of AI models used for SHM and on standardized metrics for evaluating the performance of AI models. Section 7 depicts the discussion, which interprets the results obtained, limitations, and future research directions. Section 8 presents the conclusions of the paper.

2. Methodology

The proposed methodology aims to achieve the following objectives:

- Identification of sensors used in SHM;
- Synthesis of sensing technologies used in SHM;
- Implementation of a mixed methodological approach, combining a statistical analysis of scientific literature with a thematic analysis of how AI processes sensory data;
- Comparison between traditional structural monitoring methods and modern AI-augmented solutions;
- Synthesis of the over-research of certain AI models, their advantages, research gaps, limitations, and contexts of applicability for each approach.

The research timeframe corresponds to the period 1 January 2020–31 December 2025, to provide an updated perspective on recent trends and emerging innovations. For this study, three internationally recognized databases were selected: WoS Core Collection, Scopus, and IEEE Xplore Digital Library. The justification for this strategic choice is based on the quality and specialization of each source. The WoS database is often considered a gold standard in the scientific community due to its rigorous selection of indexed journals. The time period analyzed is limited to 1 January 2020 and 31 December 2025 to capture

the stage of AI algorithm maturity in connection with the SHM applications. Studies prior to 1 January 2020 are analyzed for the theoretical context. The objective of this review is to identify trends associated with modern AI technologies. The selected interval strategically highlights recent innovations. The analyzed period places the middle of 2022 as the year of the emergence of large language models for the general public. This moment corresponds to an expansion regarding the integration of AI models into a wide range of applications, as well as researchers' interest in integrating them at the application level across different domains.

The journals indexed in WoS and Scopus cover a wide range of disciplines, including structural engineering, civil engineering, materials science, AI, mathematics, algorithms, and reference areas considered transdisciplinary for the research topic. IEEE Xplore is also a central source of information for the fields of electrical engineering, electronics, computing, and communication techniques. The field of SHM is considered transdisciplinary due to the development of contemporary sensors (such as FBG), data collection technologies (Internet of Things—IoT), and AI algorithms. All these areas are covered in IEEE Xplore publications.

Combining these three databases allows the review to benefit from multidisciplinary coverage capable of capturing the evolution of the hardware component (sensors) relative to the software component (AI algorithms) in order to guide researchers in exploring areas that were less researched at the time this material was prepared. Limiting the search to these three platforms creates data focused on works that have undergone strict peer-review processes. This methodological framework is also supported by standard practices in the field, where numerous recent reviews, even those not exclusively focused on meta-analysis, adopt systematic review principles to ensure rigor, explicitly mentioning adherence to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to guarantee the transparency of the literature selection process (check PRISMA_2020_checklist file in the Supplementary Materials Section).

The literature identification process is based on a systematic search strategy. The authors aim to provide complete coverage of the subject during the specified period, 2020–2025. The search strategy was built iteratively, starting from a general set of keywords and adapting to capture the interdisciplinarity of the areas of interest. The queries were formulated separately for WoS, Scopus, and IEEE Xplore because their indexing mechanisms and search syntax differ between the three databases. Before final processing, all raw results, regardless of their source, are imported into the bibliographic management software, EndNote, to automatically remove duplicates and ensure that each research study is considered only once in the analysis.

The search strategy was deliberately structured to capture at least two distinct categories of literature. The first research axis covers the integration of sensors into SHM in the context of construction, without imposing restrictions related to AI. The second axis explicitly focuses on AI applications in this field. This layered approach allows for the investigation of researchers' interest in the general field, and then their perspective focused on recent innovations in relation to AI.

The first search category aimed to identify a general collection of studies that use various sensory technologies in structural monitoring. The queries for WoS, Scopus, and IEEE Xplore are depicted in Figure 2. Based on the overall results, a second search category was developed that filters studies based on the use of AI. This represents the central core of the review.

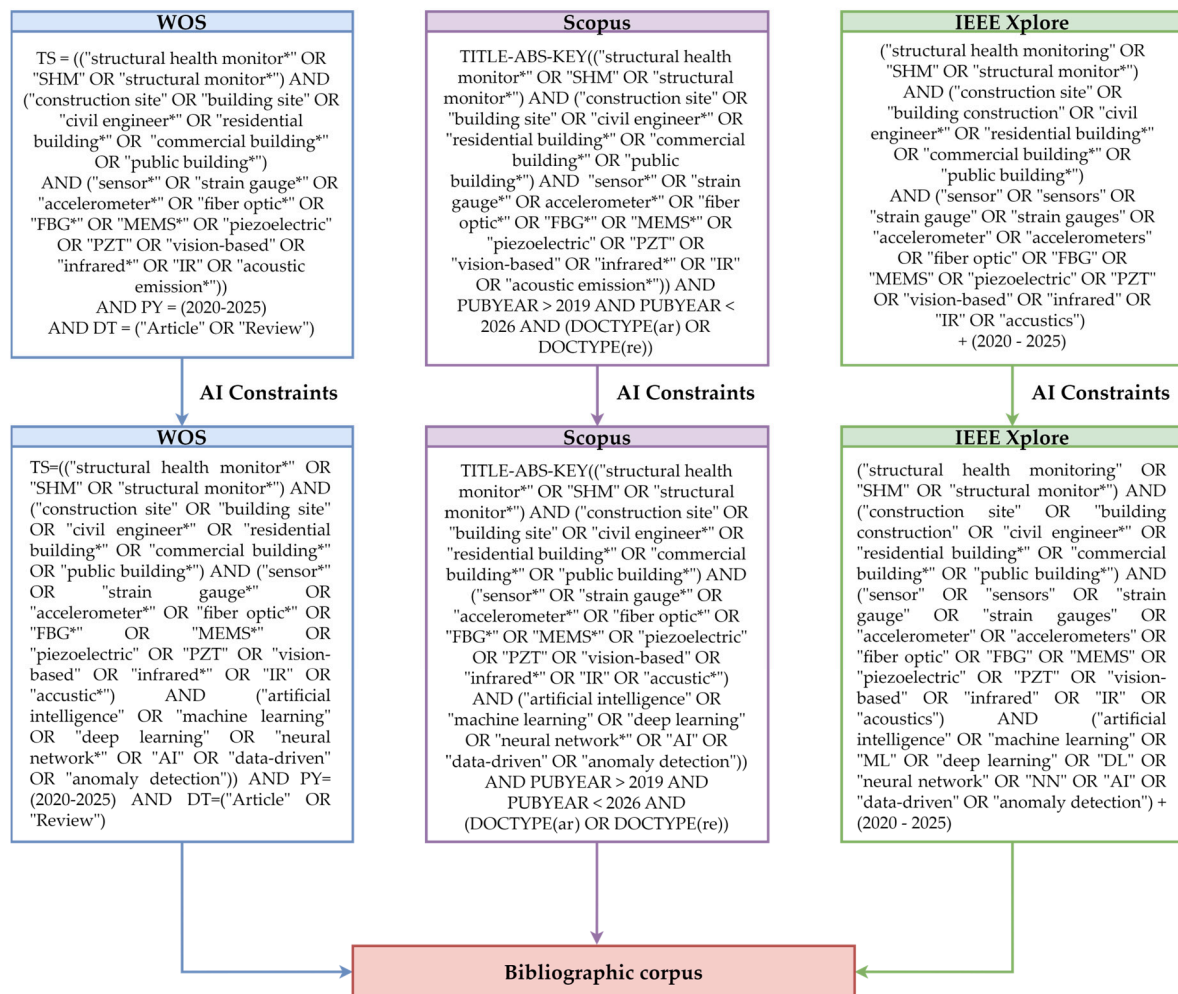


Figure 2. Scientific literature search strategy in the WoS, Scopus, and IEEE Xplore databases (the asterisk was used to include all lexical derivatives).

After obtaining all the raw results and removing duplicates, the literature selection process was organized into four successive stages in accordance with the principles of systematic reviews, and the process will be visualized through the PRISMA diagram in Figure 3.

For reproducibility, the workflow distinguishes six levels of evidence: initial database results, AI-filtered results, deduplicated records, open-access records, screened records, and the final corpus included in the qualitative synthesis. The initial database results were used to describe the general visibility of SHM and sensor research. The AI-filtered records were used to quantify the explicit connection between SHM sensors and AI. Deduplication was performed before eligibility screening so that the same study was not counted more than once in the combined analysis. The open-access restriction was then applied as a practical access criterion, and the final qualitative synthesis retained only studies that described the sensor type, AI method, SHM application, dataset, or validation setting, and at least one interpretable performance metric.

The first stage consisted of identifying in the WoS, Scopus, and IEEE Xplore the articles related to the SHM and sensors without AI concepts, with the main purpose of extracting the domain image of the research area. The basic criteria state that a document must be a research article or a scientific review, be published between 1 January 2020 and 31 December 2025, refer to the SHM of civil buildings or infrastructure, use at least one of the specified sensors, and explicitly discuss integration with AI (ML, DL, and NN).

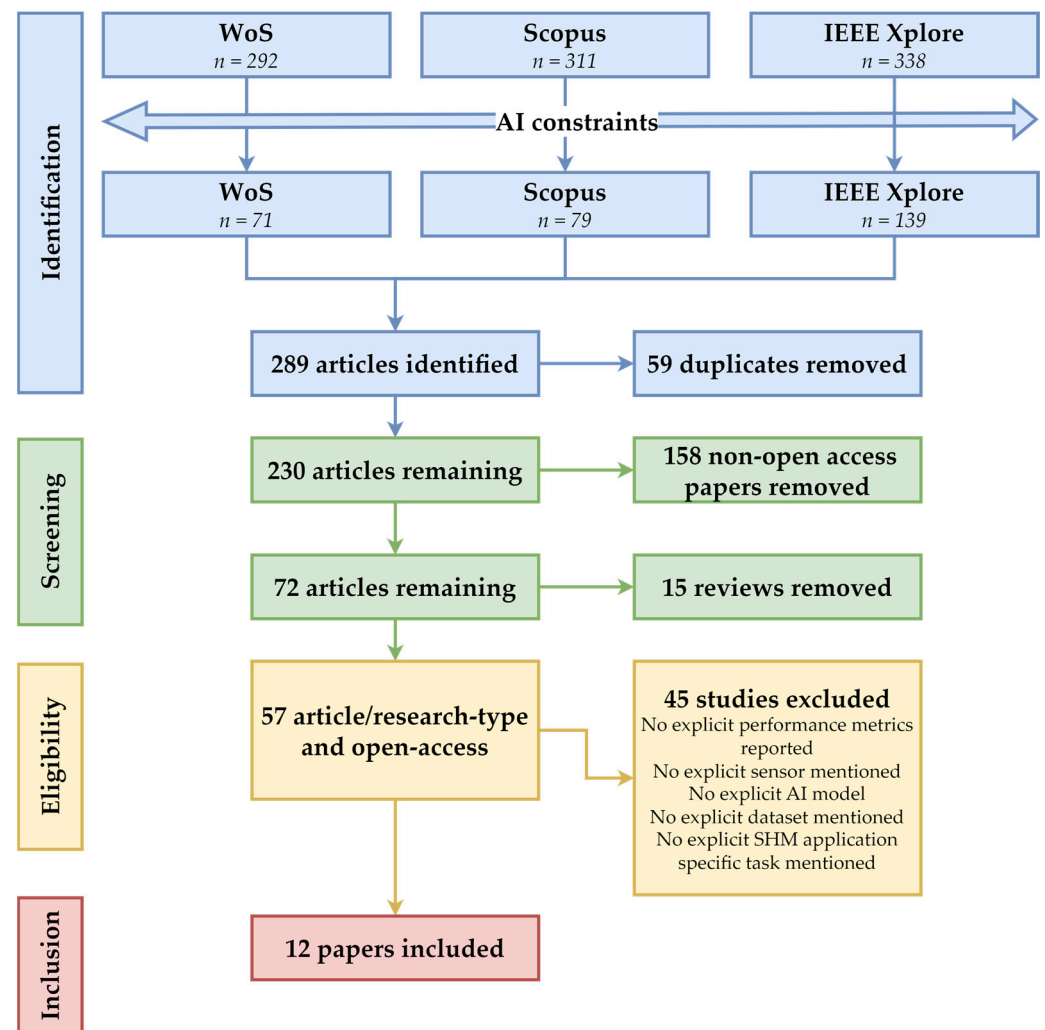


Figure 3. PRISMA diagram illustrating the article selection process.

After removing duplicates and reviews, eligibility criteria were applied, excluding studies that did not contain explicit performance metrics, explained AI models, explicitly mentioned datasets, or specific SHM applications. From the 72 articles remaining after the eligibility assessment, 57 articles or research-type open-access papers were selected for full-text inspection, of which 26 were excluded due to a lack of information regarding sensors or AI methods. Finally, 31 articles were included in the detailed qualitative analysis. The performance synthesis was further restricted to the records that simultaneously reported the SHM task, sensor type, AI model, validation context, and comparable metrics; this explains why some tables analyze a smaller methodological core than the broader qualitative corpus.

The 31-study corpus supports the qualitative synthesis, including the taxonomy of sensors, AI model families, SHM tasks, validation contexts, limitations, and research directions. Within this corpus, the 12-study methodological core (Appendix A) is a stricter subset used only for performance-oriented comparisons in the two tables that summarize supervised, unsupervised, and hybrid AI methods for SHM applications described in Section 6, because these studies reported all variables needed for task-specific interpretation. Consequently, conclusions about field trends draw on the 31-study corpus, while statements about reported model performance are limited to the 12-study core and are interpreted through the evidence-confidence framework.

To strengthen the selection pathway, Table 2 displays the reported counts. The 31 studies represent the qualitative corpus, while the smaller metric-comparable core includes only studies that simultaneously reported the SHM task, sensor modality, AI model, validation context, and interpretable performance indicators.

Table 2. Literature selection and filtering workflow.

Filtering Stage	Records In	Records Out	Exclusion Criteria/Rationale	Cumulative Retained
Stage 1. Initial database search (SHM + sensors, no AI filter)	941 records (WoS: 292; Scopus: 311; IEEE Xplore: 338)	NA	Raw results retrieved from three databases using the first search query (Figure 2). No filtering applied at this stage.	941
Stage 2. AI-specific filter applied	941	652 records excluded	The second query introduced AI-related terminology (ML, DL, NN). Retained records explicitly integrating AI with SHM sensors. AI-filtered corpus: WoS 71; Scopus 79; IEEE Xplore 139	289
Stage 3. Deduplication across databases	289	59 records removed as duplicates	Duplicates identified and removed so each study is counted only once in the combined analysis.	230 unique records
Stage 4. Open access eligibility filter	230	158 records excluded	Only open-access papers were retained to ensure full-text verifiability and reproducibility. Subscription-based articles were excluded at this stage.	72 open-access records
Stage 5. Article/research-type eligibility filter	72	15 records excluded	Only article/research-type papers were retained to ensure the presence of research performance metrics	57 articles/research-type papers
Stage 6. Full-text screening and quality assessment	57	26 records excluded	Full-text inspection applied. Excluded studies lacking: (a) explicit performance metrics; (b) identified AI model; (c) described dataset or validation setting; (d) specific SHM application.	31 studies included in qualitative synthesis
Stage 7. Metric-reporting subset (performance tables)	31	19 records excluded from tables	Further restricted to studies simultaneously reporting: SHM task + sensor type + AI model + validation context + comparable metrics. Used for the two tables that summarize supervised, unsupervised, and hybrid AI methods for SHM applications described in Section 6.	12 studies (WoS methodological core)

The possible effect of the open-access restriction was assessed by comparing the final corpus with the broader AI-filtered retrieval at the metadata level. The qualitative corpus represents 43.1% of the 72 unique eligible candidates and 59.6% of the 57 article/research-type and open-access full-text papers. The same main directions remain visible: vibration and acceleration data support detection and classification; image, video, LiDAR, and point-cloud data support visible-defect localization; FBG, strain, and PZT sensing support deformation and early-damage monitoring; and recent studies increasingly discuss physics-informed learning, digital twins, uncertainty, and edge/IoT deployment (Table 3).

To further reduce the uncertainty associated with the open access restriction, the qualitative corpus was benchmarked against the wider eligible retrieval and against high-impact subscription-based studies retained as contextual references. This comparison was performed at the level of publication venue, sensor modality, AI model family, SHM task, and validation setting. The benchmark showed that the dominant directions identified in

the included corpus, namely vibration-based monitoring, vision-based defect detection, fiber-optic sensing, supervised/deep-learning models, and the emerging use of physics-informed learning and digital twins, are also present in the broader literature. However, subscription-based studies are acknowledged as potentially more visible in bridge-scale operational monitoring, long-term field deployment, and advanced transfer-learning workflows. Therefore, the open access corpus is treated as sufficiently representative for mapping the main sensor-AI patterns, but not as an exhaustive basis for estimating the prevalence of all methodological developments in the wider SHM-AI field.

Table 3. Representativeness check for the open access qualitative corpus.

Dimension	Broader AI-Filtered Pattern	Pattern retained in Final Corpus
Vibration SHM	Accelerometers and dynamic features are used for detection/classification.	Retained in qualitative and performance synthesis.
Visual SHM	Images, video, UAVs, LiDAR, and point clouds are used for visible damage.	Retained for crack, defect, and deformation analysis.
Strain/PZT/FBG	Used for deformation, localization, and incipient defects.	Retained, including limits for very small defects.
Hybrid deployment	Physics-informed ML, digital twins, uncertainty, edge/IoT.	Retained in discussion, limitations, and future directions.

Once the final studies are selected, the data is extracted into tables dedicated to qualitative analysis. These tables aim to use the same sensors, different AI models, but common performance metrics, to identify the best results reported in the literature. This selection process ensures that the review conclusions are based on a relevant set of scientific evidence. The qualitative analysis was constrained by the requirement for open-access articles in order to support novelty, transparency, verifiability, and reproducibility. However, open access is treated here as a practical limitation rather than as a scientific quality criterion. This criterion was introduced by the authors for the detailed analysis. If the paper is not free, it is impossible to access such a large number of papers and pay for all the articles. In the author's opinion, a valuable paper is published as open-access, as it needs to be read by a consistent number of researchers. This decision may introduce selection bias because influential subscription-based studies from IEEE, Elsevier, or other publishers may be underrepresented. Therefore, the conclusions are interpreted as representative of the accessible and fully verifiable corpus, not of the entire SHM-AI literature.

This scientific review adopts a mixed methodological approach. The strategy used combines quantitative, data-driven analysis with the depth of qualitative, performance-oriented analysis. Quantitative analysis serves to describe and macroscopically synthesize the research field, highlighting researchers' interests, research trends, the distribution, and the overall vision of the literature regarding contemporary technological developments. Qualitative analysis interprets the content of individual studies by exploring the relationships between different concepts, such as sensor type, AI algorithm, diagnostic task, and reported performance metrics.

This research distinguishes itself from other review articles through its explicit comparative analysis between traditional SHM approaches and modern AI-augmented solutions. This comparison aims to be an analysis that seeks to highlight the advantages, disadvan-

tages, models used, quality reporting metrics, implications, and ideal application contexts for each paradigm. This analysis offers a pragmatic perspective on the field's evolution.

Traditional SHM methods are predominantly based on physical models and mathematical equations that describe the structure's behavior. These are based on an understanding of structural mechanics terminology and structural dynamics [29]. Classical techniques such as Experimental Modal Analysis (EMA) employ controlled excitations and sensor-measured responses to extract the structure's modal parameters, such as natural frequencies, mode shapes, and damping coefficients [30]. The advantage of these model-based methods is their ability to provide a mechanical understanding of the observed phenomena, making them accepted in traditional engineering practice [31]. However, these methods face some limitations. Firstly, modeling a real structure, especially a complex construction, during execution, is computationally extremely expensive and requires structural modeling experts [32,33]. Traditional models ignore non-linear effects, progressive degradation, and the influence of environmental factors like temperature and humidity [34]. These behaviors mask the damage signal. These methods are difficult to scale to legacy and diverse infrastructure, where data collection and model validation become a huge task [35]. The dependence on human expertise in interpreting the results makes them susceptible to judgment errors [31]. As a consequence, it is difficult to standardize.

In contrast, AI-based approaches represent an alternative, data-dominant paradigm that learns directly from raw sensor data without requiring an explicit physical model of the structure [36,37]. These methods (ML, DL, NN, etc.) are capable of identifying non-linear patterns in sensory data, thus overcoming the limitations of traditional models' simplifying assumptions [38,39]. ML models process massive volumes of data from sensors distributed throughout an entire structure. It enables real-time monitoring. As a consequence, it provides an overview of the structure's condition [40]. AI models, especially unsupervised ones, also learn the normal behavior of a structure. In return, the models detect previously unknown anomalies without being explicitly trained with examples of damage [41]. This makes them useful for identifying unconventional defects. The advantage is the reduction in dependence on human expertise in the analysis stages. Automating the feature extraction and diagnostic process makes AI technology accessible to engineers who are not specialists in SHM [35]. Additionally, AI models are superior in prediction tasks. An example in this area is the estimation of the remaining life of a structure. This is possible due to their ability to model complex relationships between input data (loads, environment) and output (degradation state) [23].

The combination of these two approaches leads to the concept of Physics-Informed Machine Learning (PIML) [38]. AI models are constrained by the principles of physics. Their predictions are realistic. Traditional models are used to validate the results of AI models. They provide a physical reference point. On the other hand, AI models are used to correct traditional models, for example, to model the non-linear or environmental effects that they ignore. The concept of a "Digital Twin" is an example of this interdisciplinary combination. A physical digital model of a structure is continuously fed with real-time data from sensors. They are adjusted by AI algorithms to accurately reflect the current state. They also analyze the future behavior of the physical structure [42–44]. As a result, the revision combines traditional physical modeling with AI learning.

In this paper, hybrid AI refers to more than the simple use of multiple learning algorithms. The term includes combinations between ML/DL architectures and structural mechanics knowledge, finite-element-informed learning, model updating, physics-informed constraints, Bayesian uncertainty quantification, and Digital Twin updating. Conventional ML/DL combinations are therefore interpreted as algorithmic hybrids, whereas physics-informed and digital-twin-based approaches are treated as structural-engineering hybrids

because they constrain or update the AI output using the behavior expected from the physical system.

3. Quantity Analysis

This section focuses on the WoS database for bibliometric visualization because it provides a consistent metadata structure for temporal, category, and keyword network analyses. WoS yielded 292 initial papers, and refining the search criteria by introducing AI-specific terminology with sensor integration generated 71 papers that constitute the bibliometric corpus used for temporal distributions, WoS categories, keyword networks, and researchers' interest in the analyzed field over time. From this WoS set, 27 open-access articles were individually analyzed to extract performance metrics reported in the fully accessible literature. In the final stage, 12 WoS articles simultaneously met all metric-reporting requirements and were included in the WoS performance tables. The broader qualitative synthesis of the manuscript, however, is based on the combined screening logic applied across WoS, Scopus, and IEEE Xplore, which led to 31 detailed studies.

The rigor of the analysis derives from the strict criteria applied to the specialized literature. Consequently, a limited number of articles were obtained that include the constraints regarding the explicit integration of sensors with AI models, the reporting of performance metrics, the description of SHM applications, and the availability of full content access. The field of SHM and AI is extensive, but many publications address components separately or do not provide methodological details for comparative analysis. For these reasons, after applying all the constraints, 12 studies were obtained, represented by the methodological core proposed in this research.

The temporal distribution of publications shows an increase in interest in recent years. Most works were published in 2025, when 24 articles were reported. Figure 4 shows that there were 16 articles in 2024, 10 in 2023, 9 in 2022, 6 in 2021, and 6 in 2020. The trend indicates that the use of AI in SHM for construction is an expanding field, supported by the maturation of AI algorithms. On the other hand, the increase in processing capacity and dense sensor networks encourages the development of this subfield.

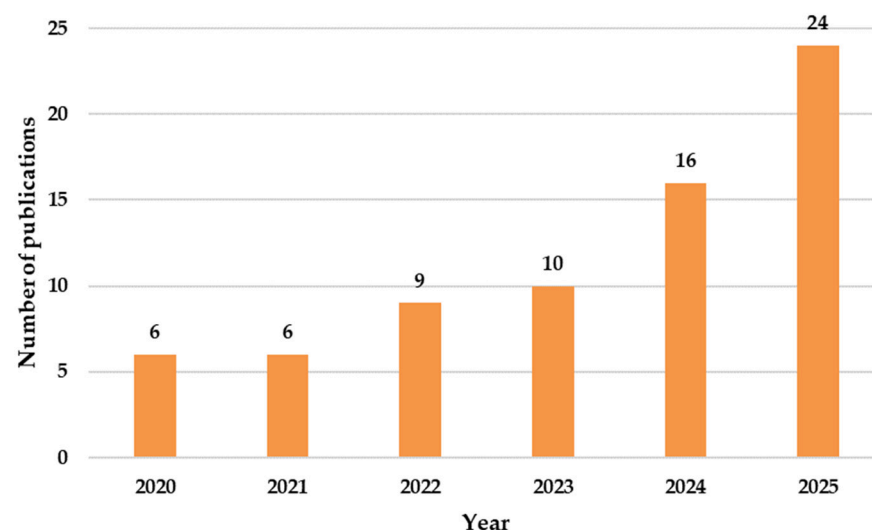


Figure 4. Temporal distribution of AI-assisted SHM publications in WoS (2020–2025).

Figure 5 shows the distribution of articles by WoS categories. Initially, a preference for Civil Engineering is observed, with 25 articles reported, followed by Instrumentation with 20, Multidisciplinary Engineering with 18, and Electrical and Electronic Engineering with 13. The distribution shows that AI-assisted SHM is not a strictly civil engineering

problem, but rather involves interdisciplinary development. The presence of the Computer Science, Artificial Intelligence, and Interdisciplinary Application categories indicates that AI-specific algorithms for sensor data analysis are an intensively researched direction within the context of SHM. On the other hand, the Materials Science, Applied Physics, and Chemistry Analytical categories show interest in monitoring material properties in the context of developing sensors that detect the physicochemical degradation of structures. Disciplinary diversity is correlated with modern SHM solutions that require an integrated hardware-software approach.

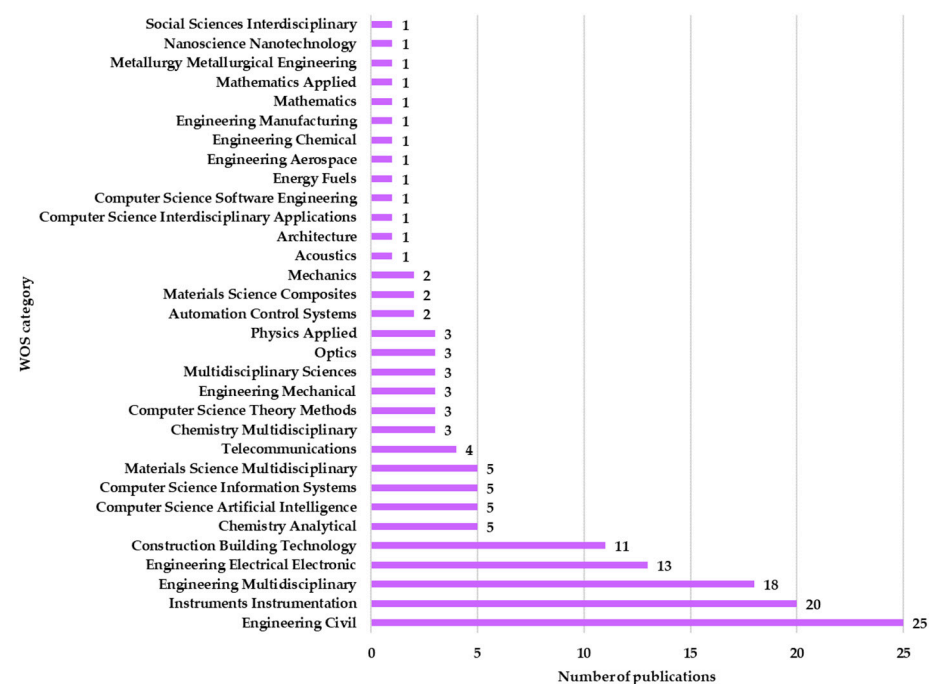


Figure 5. Distribution of articles by WoS category (2020–2025).

Figure 6 presents the keyword network map obtained using the VOSViewer 1.6.20 tool. To achieve this, a minimum of five appearances per keyword was required for the 71 articles, resulting in four clusters. The central node of the network is represented by the term SHM. This demonstrates that literature treats SHM as an integrated system. In this ecosystem, sensor data is analyzed using advanced AI methods, as the other clusters relate to ML, DL, and NN. The clusters reflect the transition from traditional methods based on physical models to data-driven approaches. They handle large volumes of data, non-linear relationships, behaviors that are difficult for humans to process, and high processing speed. The distinct presence of the term “Anomaly Detection” highlights the growing interest of the academic community in unsupervised methods, which are used in situations where data is insufficient or unlabeled, a frequent occurrence in real-time acquisition applications. A bibliometric analysis of the keywords presented in Figure 6 shows the interconnection between sensor development, SHM applications, AI methods, and civil engineering, with applications in integrated SHM systems.

In Figure 6, the four clusters marked in red, yellow, green, and blue are presented. The red cluster includes the terminology SHM, AI, Anomaly Detection, and Civil Engineering. This cluster is the conceptual core of the field because it integrates specific SHM concepts with AI in an applied context. The green cluster includes SHM, Sensors, NN, and Damage Detection as terminology. This cluster highlights the connection between sensor infrastructure and NN models used in identifying structural degradations. The blue cluster is centered on specific AI techniques, such as DL and Computer Vision, in relation

to Crack Detection. This approach shows a focus in the field on identifying the orientation toward visual applications and advanced image-level analysis models in the direction of surface defect detection. The yellow cluster contains the terminology Localization and ML. This cluster shows the distinct direction oriented toward identifying defect positions through ML methods applied to vibrational or multisensory data. Figure 6 shows the existence of the four directions that bring together the conceptual integration of AI-SHM.

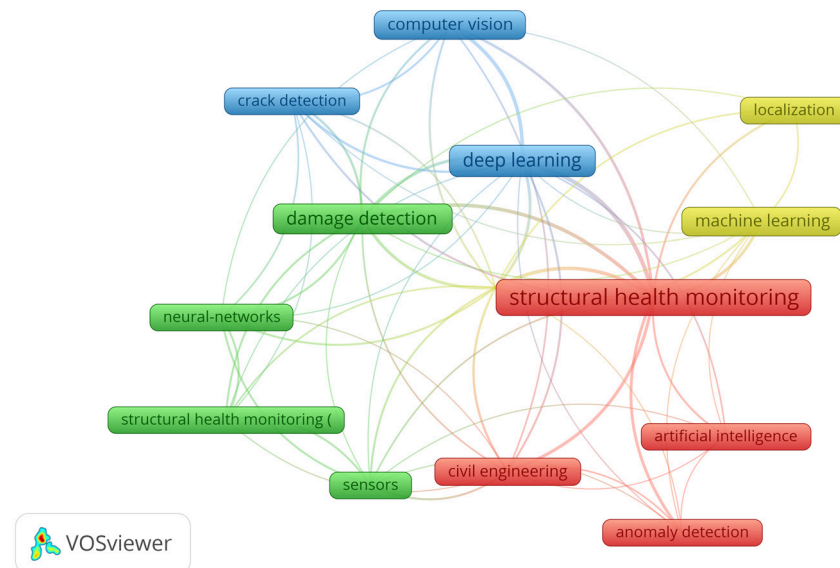


Figure 6. Keyword network for WoS articles analyzed with the VOSViewer tool (2020–2025).

Figure 7 shows the distribution of articles by publisher. In this representation, it can be observed that most articles are disseminated in Elsevier, followed by MDPI, Sage, IEEE, and Wiley. The dominance of Elsevier and MDPI is explained by their open-access policies and the journals' interest in interdisciplinary research that includes AI applications. These are established journals in structural engineering and materials science, particularly through their openness to modern technologies such as AI tools. These reputable publications also offer a high degree of scientific validation for the reported results.

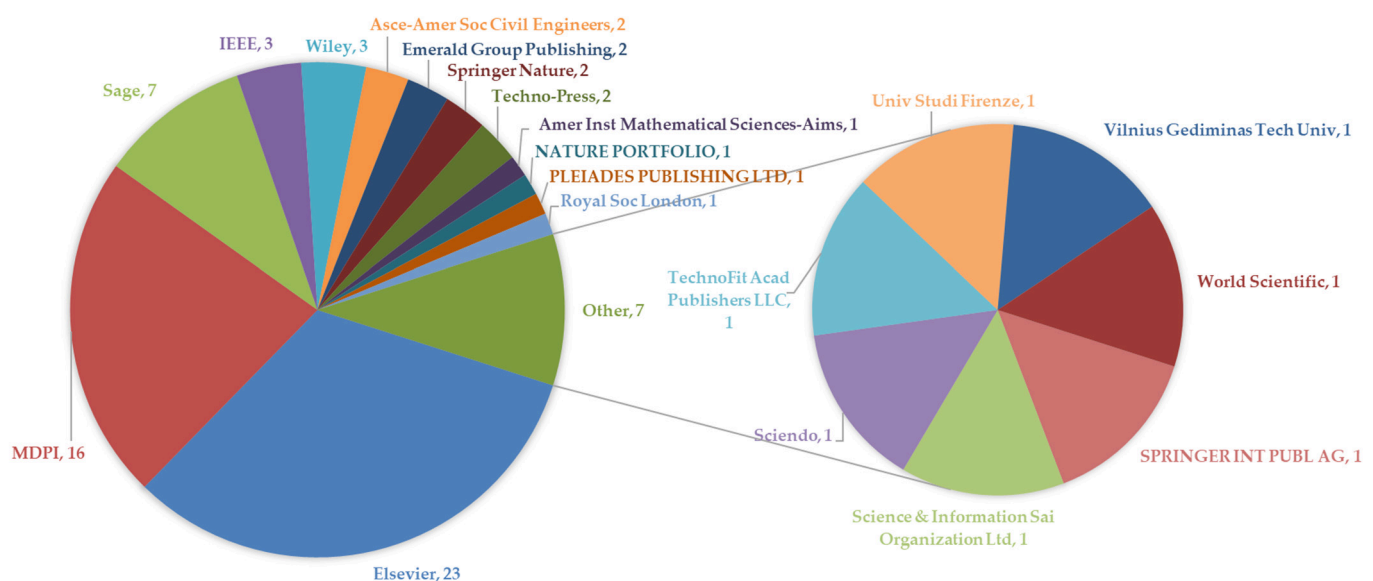


Figure 7. Distribution of articles by publisher (2020–2025).

The quantitative analysis of the IEEE Xplore results shows 338 results for the first query (Figure 2), of which 198 are in conferences, 130 in journals, 7 books, and 3 magazines. The distribution shows the engineering community's interest in the rapid dissemination of results through conferences. The second query (Figure 2), which explicitly includes AI-specific terminology, yielded 139 results, including 82 conference papers, 54 journal articles, 2 books, and 1 magazine. Furthermore, a preference for disseminating results in conference papers can be observed. After applying the open-access constraint, only 19 papers resulted, which limits the possibility of conducting a qualitative analysis exclusively within the IEEE Xplore database.

Quantitative analysis demonstrated through numerical values that the AI-assisted SHM field for construction is in a growth phase, and it is characterized by interdisciplinarity and a data-driven approach. Limitations on open access, particularly in IEEE Xplore, indicate the need for more open policies to facilitate technology transfer and the reproducibility of results.

4. SHM Application Taxonomy

The taxonomy of SHM applications is centered around monitoring applications. Scientific literature traditionally defines four categories of tasks in SHM: damage detection, localization, classification, and prognosis [45], as depicted in Figure 8.

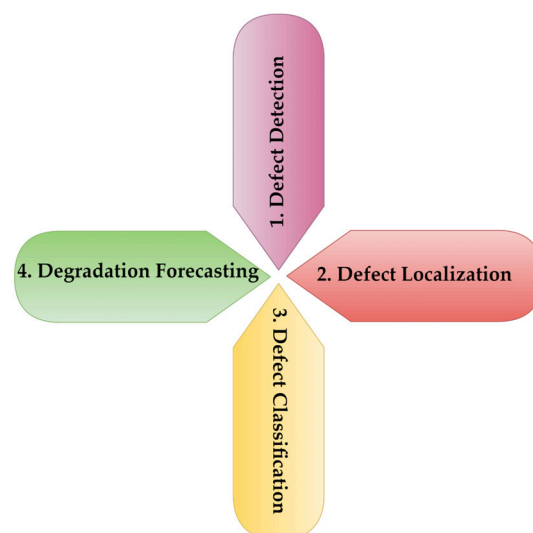


Figure 8. Taxonomy of SHM applications: detection, localization, classification, and prognosis.

The first task analyzed is **Defect Detection**. The purpose of this application is to determine whether a defect or alteration in structural integrity is present in a component of the structure [45]. A defect is defined by Ahrens and Hollander [2] as any change in the material properties, topology, geometry, or connectivity of a structural system. The defect affects its performance, without necessarily leading to a complete loss of function [2]. Detection is achieved by analyzing changes in the structure's input-output signals. Examples of changes include deformations, accelerations, and stresses, and they are recorded by sensors. For example, a change in the natural frequencies or mode shapes of a structure indicates the presence of a crack or corrosion [46]. The detection stage is important because all subsequent tasks depend on the validation of its existence. Detection methods can be supervised or unsupervised. The first is defined as models that are trained on data labeled "defective" or "non-defective". Unsupervised learning describes algorithms with the ability to identify anomalies in the input data without prior labels [47].

After confirming the existence of a defect, the next task is **Defect Localization**. This involves determining the exact position or approximate area where the damage is located [45]. The location coincides with the planning of maintenance and repair interventions. Localization methods use information from sensors distributed throughout the structure to triangulate or reconstruct the positions of abnormality sources. For example, in the case of FBG sensors, the location can be identified relative to the sensor that detected the largest change. In the case of acoustic or ultrasonic-based sensors, the wave propagation time from the defect location to multiple sensors can be used to calculate its position [48]. Some AI methods, such as convolutional neural networks (CNNs), are trained to locate defects in images of the structure or even in raw sensor data. These produce probability maps of defect locations [23].

Once a defect has been detected and located, the next step is **Defect Classification**. This task involves characterizing the type of defect (e.g., crack, erosion, corrosion, delamination) and, in many cases, its severity [45]. The classification assesses the actual impact of the defect on structural integrity and safety. For example, a small superficial crack has far less serious consequences than a deep fissure in a stress concentration area. ML models are used in classification tasks. Raw sensor data (vibration signals, images, stress data) are transformed into representative features (feature extraction), which are then used to train classifiers such as Support Vector Machines (SVMs), Decision Trees, or Deep Neural Networks (DNNs) [15,46]. A drawback in classification algorithms is the handling of variable conditions, such as temperature or load fluctuations, which can mask or mimic fault signals. Therefore, advanced domain adaptation methods are often necessary to ensure the model's classification accuracy [49].

The final task in taxonomy is **Degradation Forecasting**, sometimes also referred to as damage forecasting [50,51]. In contrast to the other three tasks, which focus on the current or past state of the structure, prognostication aims to estimate the future behavior of the defect [52]. Objectives include estimating the defect growth rate and the remaining time until a critical state or component failure [53]. This is a natural extension of detection and localization and is necessary for implementing predictive maintenance strategies and long-term risk management [51,54].

Prognosis is linked to the deterministic or ML models [55]. The last one types are trained on chronological monitoring data to learn defect evolution trends. Regression models are used to make temporal predictions [56,57]. For example, such a model is trained to predict the future length of a crack based on its current length and the history of load cycles. The performance of these models is evaluated using a combination of regression metrics.

Integrating prognostics within SHM frameworks activates a closed-loop cycle. Future predictions affect maintenance decisions. In return, it reduces the degradation rate, thus closing the structural health management loop [58]. This hierarchy of tasks—detection, localization, classification, prognostication—forms a framework for the development of complete SHM solutions.

Defect detection and localization are the first steps in the SHM process for buildings. The success of these tasks is correlated with the accuracy of classification and prediction. In recent years, AI-based methods have demonstrated their ability to extract weak degradation signals from noisy data collected by sensors. These techniques are applied to data from a wide range of sensors.

Figure 9 presents the SHM workflow integrating physical sensors and AI algorithms, along with a taxonomy of applications. The SHM workflow follows the AI applications data processing steps: (1) data collection with specialized sensors, (2) signal processing, (3) feature extraction, (4) training for the AI models, and (5) decision support for AI

model validation. The application taxonomy categorizes the included studies based on civil structures, sensor technologies, AI methods, and monitoring purposes.

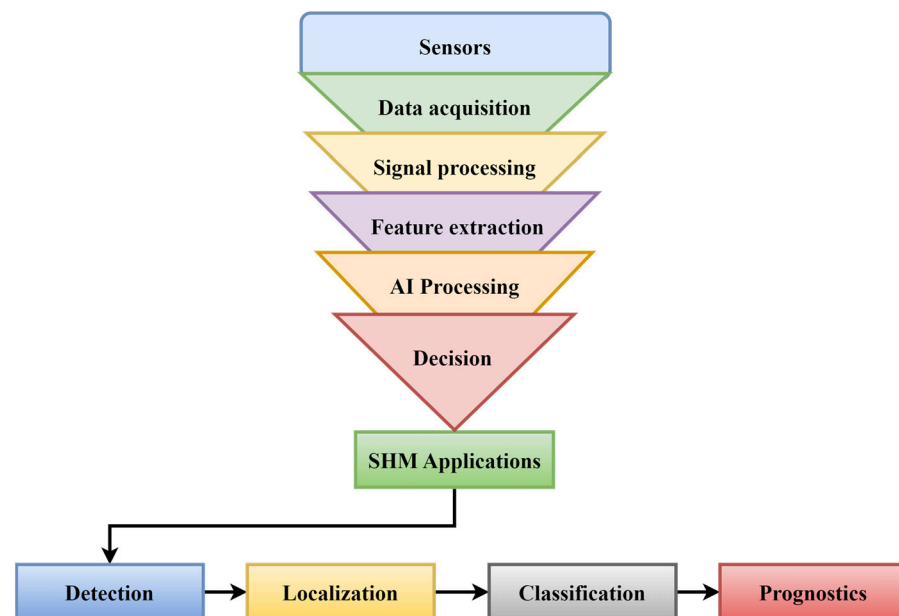


Figure 9. SHM workflow integrating physical sensors and AI algorithms.

The flow presented in Figure 9 outlines the general stages of the AI model training process, which may exhibit certain particularities depending on the type of learning used. Supervised methods require labeled data during the training stage in order to perform training. Subsequently, validation on predefined defect classes is a step that cannot be omitted in the AI model development process. For unsupervised models, learning the normal behavior of the structure and identifying deviations is done without explicit labels. Hybrid models combine physical constraints or traditional models with AI architectures, forming parallel analysis pathways.

Figure 9 presents the authors' proposal for the SHM workflow. The vertical section illustrates the sequential stages of the physical element process up to the AI component, which converge toward SHM Applications. The lower part highlights the four stages of taxonomy: Defect detection, their location, classification of damage types, and prognosis of structural condition. The diagram illustrates the conceptual approach for integrating physical sensors with AI algorithms to assess the condition of civil structures through real-time data analysis. Figure 9 provides a systematic overview of the monitoring process and its practical applications in modern civil engineering.

5. AI Models Used for SHM

A research area in the specialized literature is vibration-based SHM. Sensors such as accelerometers and PZT sensors are used to measure the dynamic response of the structure to natural or artificial loads [59]. The basic idea is that structural damage (e.g., cracking, corrosion) leads to changes in its mass properties, stiffness, or damping. ML methods are then used to detect and locate these changes. For example, models such as Random Forest (RF), Support Vector Machines (SVM), or Artificial Neural Networks (ANN) are trained to classify the structural condition based on features extracted from vibration data [46]. Naresh et al. [46] compared different ML algorithms on simulated data for a rotating bridge system. The paper demonstrated the possibility of using such algorithms in detecting and locating structural changes. In some situations, these methods require manual feature extraction. This task is difficult and depends on the engineer's experience. To overcome this

limitation, DL methods, especially CNNs, are used to automatically extract features directly from raw vibration data or from visual representations of it, such as spectrograms [57,60]. Yoon et al. [48] demonstrated the use of a CNN to detect and locate cracks in real-time, using data from a small number of sensors. This result demonstrates the possibility of implementing cost-effective monitoring systems at the level of real structures.

Another research direction involves the use of OFS, particularly FBG sensors, in combination with AI methods for stress, temperature, and strain monitoring [5,6]. FBG sensors are immune to electromagnetic interference, making them suitable for structural monitoring of building blocks [5,7]. Data generated by dense FBG sensor networks is analyzed by AI models specialized in detecting anomalies. For example, an SHM system implemented for an ancient wooden building used 56 FBG sensors to monitor beam deflection and column inclination. Monitoring was carried out over the course of a year [7]. Contemporary approaches combine OFS with DL to optimize signal demodulation and manage the complexity of dense sensor networks [6]. Additionally, distributed sensing technologies like DAS (Distributed Acoustic Sensing) offer high-resolution spatial and temporal data that can be processed using DL algorithms to detect discrete events such as impacts, cuts, or intrusions along a long section of infrastructure, such as pipelines or tunnels [5].

Image-based sensors (conventional cameras and LiDAR) represent fertile ground for surface defect detection applications. They check the visible integrity of structures, identifying cracks, fissures, parquet degradation, or corrosion [11,14]. DL methods (CNNs) have become the most common standard for object detection and segmentation tasks in images within this context. Models like You Only Look Once (YOLO) or U-Net are used to automatically detect and locate cracks in images of concrete or asphalt surfaces [14,15]. The performance of these models is evaluated using standard metrics for classification and segmentation. The usual metrics are precision, recall, F1-score, and mean average precision (mAP) [61,62]. For example, a study on crack detection reported an accuracy over 99% and an Area Under the Curve-Receiver Operating Characteristic (AUC-ROC) of 99.95% [63,64].

Zinno et al. [65] analyze the way AI techniques are involved in SHM at the bridge level. The applications identified in the paper aim at detecting and locating damage through vibrational analysis and classification models. It also aims at pattern recognition using ML algorithms, integrating data from IoT sensors for continuous monitoring and real-time evaluation, optimizing sensor placement to reduce measurement uncertainties, and predicting damage progression through forecasting models. The conclusions of the research [65] refer to AI-based SHM systems in relation to reducing maintenance costs and increasing infrastructure safety. The paper mentions that the installation of sensors, management of large data volumes, noise filtering, and selection of appropriate analytical frameworks remain reference elements for the SHM field in relation to IoT-AI. The paper by Sabato et al. [66] explores the evolution of non-contact techniques used in SHM monitoring. The authors find that these methods are expensive, time-consuming, and provide information only at the points where sensors are installed. Furthermore, the authors [66] also identify SHM applications such as measuring displacements and deformations, infrared thermography for detecting internal defects and areas with material loss, LiDAR for terrestrial reconstructions and monitoring crack evolution, AI-assisted video analysis for vibrations, modal analysis, and automatic anomaly detection, as well as integration with AI algorithms for damage classification, image segmentation, and defect evolution prediction. Harle et al. [67] synthesize how AI and advanced data analysis have implications in SHM. The authors find in their research that their integration allows for the processing of large volumes of structural data, leading to rapid detection and diagnosis of damage progression. Additionally, AI methods improve accuracy and anomaly detection, allowing a transition from reactive maintenance to predictive maintenance.

Vijayan et al. [20] highlight a novel element in the specialized literature, which refers to the integration of IoT infrastructure with AI models and non-destructive testing in the transformation of SHM through continuous monitoring for early damage detection, predictive analysis, and maintenance optimization. In this way, the authors [20] make a transition from manual inspections to connected intelligent systems. This transition enhances safety, durability, sustainability, and trust in IoT infrastructures that reduce costs, increase accuracy, and eliminate risks associated with traditional inspections using AI technologies. The research by Dinh et al. [68] analyzes the evolution of structural monitoring at the railroad level. In this paper, traditional inspections that can be migrated to advanced sensor and AI-based systems are also addressed. The research shows that the dominant themes focus on the development of data acquisition technologies for defect detection using AI models through predictive maintenance models. Sun et al. [25] emphasize the importance of optimizing sensor placement in SHM monitoring. This aspect arises from the fact that their positioning directly correlates with the accuracy of the measurements, costs, and system performance.

Table 4 presents results from the specialized literature regarding the sensor used in the research, the AI model employed, the measured parameters, and the specific application.

Table 4. Correlation of sensor types, AI models, and SHM applications.

Sensor	AI Method (ML/DL/NN)	Measured Parameters	Specific Application	Reference
Accelerometers	CNN	Dynamic response	Damage detection and classification in vibrating structures	Naresh et al. [46]
Accelerometers	CNN and Transfer Learning	Acceleration response histories; compressed histograms; mean, standard deviation, scale factor	SHM: damage identification and localization in numerical and experimental vibrating structures	Azimi and Pekcan [60]
Vision-based (camera images)	CNN	Crack images, crack geometry	Automated crack detection in civil infrastructure	Shao et al. [69]
Accelerometers (vibration sensors)	CNN	Vibration signals; acceleration time series	Vibration-based damage detection; fast inference for embedded SHM	Wang et al. [70]
WSN nodes	DL	Time-series structural response	prediction of SHM state	Liu [71]
LM35 embedded temperature sensors (IoT-based SHM)	ANN	Hydration temperature, TTF, and early-age compressive strength	Prediction of early-age concrete strength; non-destructive SHM using IoT and ANN	Palanisamy et al. [72]
Accelerometers (dynamic response sensors on ASCE benchmark structure)	DL–1D CNN	TFs derived from acceleration signals; frequency-domain response ratios	Vibration-based structural damage identification using TF-1D CNN; detection of stiffness loss in braces under random excitation	Liu et al. [73]
Vision-based sensors (RGB cameras, 3D point-cloud acquisition systems)	DL–3D CNNs, segmentation networks, object detection models	3D geometry, point clouds, structural surface features, crack morphology	Generation of synthetic 3D datasets for training DL models in SHM; improved crack detection and structural inspection	Shao et al. [74]
Accelerometers	NN	Vibration time-series; compressed CNN features; spatial-temporal correlations; graph-based embeddings	High-accuracy structural damage detection and localization under noise	Wijethunga et al. [75]

Note: 1D—One-dimensional; RGB—red, green, blue; TF—Transmissibility function; TTF— Temperature–time factor; WSN—Wireless Sensor Network.

The studies presented in Table 4 highlight the accelerated evolution of SHM in terms of both sensor technologies and the use of AI for detecting and assessing structural damage. Visual monitoring is inspected using RGB cameras and DL techniques that allow for automatic crack detection, as shown in the crack detection study [69]. In parallel, FBG sensors are analyzed in review articles, highlighting their ability to measure strain, temperature, and humidity in concrete structures or composite materials [16,19].

Reparameterized CNNs are used to accelerate inference in degradation detection [70], and hybrid techniques like Variational Mode Decomposition–Gated Recurrent (VMD-GRU) optimize monitoring through wireless sensor networks [71]. Other works explore the use of transmissibility functions combined with 1D-CNNs for damage identification [73].

Graph neural models integrate spatial and temporal information for damage localization [75]. Shao et al. [74] present the generation of synthetic 3D datasets that train DL models in the absence of real data. Predicting concrete properties at early ages using IoT sensors and ANNs offers a non-destructive alternative to traditional tests [72], while unsupervised methods based on autoencoders allow for anomaly detection without labeled data [76,77].

The matrix presented in Table 5 synthesizes the literature for the task-specific relation between sensory modalities and SHM applications. Detection, being the simplest task, works with the help of an extensive class of sensors. Localization must be interpreted according to the physical information provided by each sensor. Image- and video-based methods can provide surface or geometric localization of visible damage through crack segmentation, object detection, DIC, LiDAR, or 3D point cloud analysis, but they do not provide wave-propagation-based localization in the same sense as PZT or acoustic-emission systems. Strain sensors provide strain-field or sensor-position localization, while vibration sensors often support global-response-based localization. Classification requires feature-rich representations: vibrations offer distinct spectral signatures, and images allow for visual recognition of defect types. Prognostics requires a continuous degradation time series; vibrations and deformations monitor the physical evolution of the structure, while images and acoustics require repeated acquisition and consistent environmental conditions to become predictive in the long term [78,79].

Table 5. SHM application matrix based on sensor data type.

SHM Task/Data	Vibration	Strain	Images	Acoustic
Detection	✓	✓	✓	✓
Localization	✓	✓	–	✓
Classification	✓	–	✓	–
Prognostics	✓	✓	–	–

Note: “✓” denotes the presence; “–” denotes the absence.

The quality of the scientific proposals disseminated in the specialized literature is assessed using specific parameters of AI models, as presented below.

6. Standardized Metrics for Evaluating the Performance of AI Models

Evaluating the performance of AI models validates their use in practical applications. Equally important, performance metrics help compare models to select the most suitable one for a specific structural monitoring task [62,80]. Reporting these metrics is considered a mandatory practice to ensure the performance, applicability, transparency, reproducibility, and reliability of the results presented in the scientific literature.

For classification tasks, where the goal is to assign a label (e.g., “no damage,” “slight damage,” “severe damage”) to a data instance (such as a vibration dataset or an image), a set

of interconnected metrics is used. Accuracy is the most frequently cited metric, representing the total percentage of correct predictions out of the total number of predictions [81]. However, accuracy is unsuitable for imbalanced datasets. Basically, if most instances belong to the negative class (“no damage”), then accuracy can no longer guarantee the correct evaluation of performance, which is why metrics like precision, recall, and F1-score are used. Precision measures how many of the positive predictions (instances labeled as “damaged”) are actually correct [82,83]. Recall (or sensitivity) measures how many of the actual positive instances were correctly identified [84]. The F1-score is the harmonic mean of precision and recall [85,86].

Regression tasks predict a continuous value (e.g., voltage level, structural deformation, or energy consumption). The performance of the regression tasks is evaluated using specific metrics such as the error between the predicted and actual values. Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) are also evaluation metrics for the regression tasks [87,88]. RMSE calculates the square root of the mean squared errors and penalizes larger errors [88]. MAE calculates the average absolute errors. It provides a mean error in the same units as the predicted variable. Another metric mentioned in the literature includes the coefficient of determination (R^2). It measures the proportion of variability in a variable that is explained by the independent variables in the model [87].

For specific object detection tasks in images, such as detecting cracks on building facades, a standard metric is mAP. This measures accuracy at different confidence levels. Essentially, it performs an overall evaluation of the model’s performance across all detected object classes [62,89]. Additionally, for applications requiring rapid decisions, such as real-time alert systems, the latency metric (the processing time needed to generate a prediction) becomes a benchmark [85]. Zhao and Jebelli [85] reported a latency of 1.62 milliseconds per frame for a roof defect detection method. Presenting a comprehensive palette of metrics demonstrates the capabilities of AI models to contribute to building safety and sustainability. The very good values of the metrics reported in the analyzed literature are largely interpreted in the context of experimental conditions. Some studies use laboratory setups, benchmark models, simulated data, or controlled datasets with a limited number of classes and stable environmental conditions. In many cases, the class distribution is balanced, and validation is performed using standard methods such as cross-validation or train-test split on data from the same experimental domain. These conditions favor the achievement of very high performance, which is not necessarily directly transferable to real infrastructures, exposed to climatic variations, environmental noise, and complex degradations. Therefore, the reported metrics are interpreted as results obtained in specific contexts.

For this reason, the metrics reported in this review are grouped by SHM task and interpreted within their validation setting. Classification metrics such as accuracy, precision, recall, and F1-score are discussed separately from object-detection metrics such as mAP and from regression metrics such as MAE, RMSE, MAPE, RRMSE, and correlation coefficients. Laboratory tests, simulated structures, benchmark datasets, synthetic images, real structures, and external validation datasets are not considered equivalent evidence levels. The review, therefore, avoids ranking models across incompatible tasks and instead identifies which sensor-AI combinations appear promising within each task-specific context.

To make the synthesis more analytical, each performance claim was interpreted through the evidence-confidence framework presented in Table 6. The framework does not pool heterogeneous metrics, because tasks, sensors, structures, datasets, and validation protocols differ substantially. Instead, it ranks the interpretive strength of each reported result according to validation realism, robustness testing, transparency, and transferability.

Table 6. Evidence-confidence framework for interpreting SHM-AI performance.

Confidence Level	Typical Evidence Profile	Interpretation
High	• Real operational structure or external (independent) validation dataset • Cross-structure or leave-one-structure-out validation protocol • Environmental variability (temperature, noise, load) was explicitly tested • Missing-data or sensor-fault robustness reported	Stronger evidence for deployable SHM-AI behavior
Moderate	• Laboratory specimen or recognized benchmark structure • Clear task, sensor, model, and metric reporting • Cross-validation or train/test split on data from the same structure • Controlled conditions acknowledged; no external dataset	Credible methodologically, but transferability still needs testing
Limited	• Simulation-only or synthetic-only data • Narrowly controlled single-structure setting • Single train/test split with no cross-validation • No noise, variability, or robustness testing reported	Proof-of-concept evidence, not operational proof
Not comparable	• Insufficient task, sensor, dataset, validation, or metric details • Metrics reported without a validation context • Heterogeneous conditions prevent meaningful cross-study comparison	Discussed narratively, excluded from performance comparison

For cross-study interpretation, the evidence-confidence assessment was operationalized as a weighted qualitative ranking rather than a direct numerical meta-analysis. Each reported result was interpreted according to four criteria: validation realism, robustness of the tested dataset or damage scenario, transferability beyond the original structure or sensor configuration, and treatment of uncertainty. Higher confidence was assigned to studies using independent test data, realistic or operational validation, explicit noise or environmental variability analysis, and transparent reporting of model uncertainty. Conversely, lower confidence was assigned to results obtained only on balanced laboratory datasets, simulated data, synthetic images, or same-domain train-test splits. This ranking was used to moderate the wording of the synthesis, so high metric values are reported as evidence of methodological potential unless they are supported by realistic validation and transferability evidence.

Table 7 presents the results obtained from the WoS corpus after applying the open access constraint, which yielded 27 results. These articles were studied individually, extracting qualitative research elements through metrics. The results show a diversity of AI approaches in structural monitoring. DL models are the most studied and have also achieved the best performance, a fact confirmed by their accuracy of up to 99.65% in multi-class damage classification based on accelerometer data. All DL methods achieved 93.96% accuracy in crack classification, integrating a modern concept of synthetic data. These types of data are generated by AI methods and are not real data. In object detection (e.g., cracks), the completed metrics (Precision, Recall, $mAP-0.5 = 0.837$) confirm the possibility of using CNN models on RGB images. Interestingly, a simple ML model, such as an SVM or an RF, achieves 99.77% accuracy in the binary classification of the retrofitting status. Thus, it can be deduced that the simplicity of models remains competitive in well-defined problems.

Table 7. Performance summary of supervised AI models for SHM applications.

SHM Task	Model Type	Key Performance Metrics	Sensor Type(s)	Confidence Level	Reference
Crack image classification	DL	Precision = 93.96% (with synthetic data)	Vision-based (RGB images)	Limited	Shao et al. [69]
Crack object detection	DL	Precision = 83.82%, Recall = 84.54%, mAP-0.5 = 0.837	Vision-based (RGB images)	Limited	Shao et al. [69]
Multi-class damage classification (5–15 classes)	DL	Accuracy up to 99.65%	Accelerometers	Moderate	Dabbous et al. [90]
Binary damage classification (retrofitting state)	ML	Accuracy = 99.77%	Accelerometers (12 units)	Moderate	Presno Vélez et al. [91]
Time-series regression	DL	MAE = 0.03, RMSE = 0.04, MAPE = 0.06	Wireless Sensor Network	Limited	Liu [71]
Vibration-based structural health monitoring	NN	Accuracy = 98%	Accelerometers (18 units)	Moderate	Wijethunga et al. [75]
Structural response reconstruction	NN	RRMSE = 0.04–2% (disp./vel./acc.)	Sparse accelerometers	Limited	Wang et al. [92]
3D displacement measurement	DL	Correlation coefficient. = 0.95–0.98; Relative error <18–20%	Monocular camera and laser	Moderate	Shao et al. [93]

Note: MAPE—Mean Absolute Percentage Error; RRMSE—Relative Root Mean Square Error.

In regression (e.g., structural response prediction), DL offers very small errors (MAE = 0.03, RMSE = 0.04), and the reconstruction of the structural response with NN has errors below 2%, even when a limited number of sensors are used. The data presented in Table 7 confirm that integrating appropriate sensors (accelerometers for dynamics, cameras for surface defects) with AI architectures produces high-performance SHM systems. In the same context, the effectiveness of these systems depends on the type of task, data quality, hardware-software infrastructure configuration, the type of AI model, and the construction context. The lack of studies on residential buildings remains a notable gap [94].

The results presented in Table 8 highlight the good performance of unsupervised ML and DL models in detecting and locating structural damage. Most studies report accuracy over 90% or an F1-score over 0.96 for medium or large defects (10–20 mm). A critical observation is the dramatic decrease in performance for small defects (2 mm), where the F1-score drops to 0.09–0.15. This suggests a fundamental limitation of FBG-based virtual sensors (408 channels). The explanation comes from spatial resolution or measurement noise [95]. This observation is particularly important because it reinforces the idea that sensitivity to micro-damage remains a challenge that needs to be intensively investigated by the contemporary academic community, especially in the context of advanced architectures. In contrast, systems with PZT transducers achieve an F1-score of 0.96–0.99 for localization and size classification [96]. Additionally, integrating DL for severity assessment (not just detection) yields precision and recall up to 0.99/0.98, demonstrating the models' ability to extract complex semantic information from sensory data [97].

Table 8. Performance of unsupervised AI and hybrid methods for SHM applications.

SHM Task	Model Type	Key Performance Metrics	Sensor Type	Confidence Level	Reference
Damage detection & localization	ML	Detection accuracy > 90%	Accelerometers	Moderate	Gunes and Gunes [98]
Damage detection & localization (small to large defects)	ML	F1-score $\approx$ 99% (10–20 mm); F1-score $\approx$ 9–15% (2 mm)	Virtual FBG-equivalent strain sensors (408)	Limited	Herrera et al. [95]
Binary anomaly detection	ML	100% detection (all scenarios)	High-grade MEMS accelerometers	Moderate	de Alteriis et al. [99]
Anomaly detection and severity assessment	DL	Precision up to 99%; Recall/F1-score up to 98%	Accelerometers	Moderate	Liu et al. [97]
Damage localization and size classification	ML	F1-score $\approx$ 96–99%	PZT transducers	Moderate	Perfetto et al. [96]

The comparison also shows that high values should not be generalized without considering train-test split design, cross-validation strategy, leave-one-structure-out validation, environmental variability, damage scenarios, sensor density, and data availability. Models evaluated on data from the same structure or on balanced laboratory datasets may overestimate transferability. Leave-one-structure-out validation, external datasets, missing-data tests, and noise-robustness analyses would increase confidence that an AI-SHM model can operate beyond the conditions in which it was trained.

Using the confidence framework, isolated peak values from balanced laboratory datasets, simulated structures, synthetic images, or same-domain splits are interpreted as methodological potential rather than operational performance guarantees. The strongest conclusions are assigned to results that combine transparent metrics with realistic validation, robustness analysis, and clear sensor-task correspondence.

From Table 8, it is evident that systems equipped with unsupervised models are effective for visible damage but require further investigation into the sensory infrastructure (e.g., higher-density, hybrid sensors) and of few-shot learning or uncertainty quantification algorithms to detect incipient defects. Identifying them is a key step in SHM processes for the preventive maintenance of old or residential buildings.

Tables 7 and 8 exclusively synthesize the results extracted from this final subset of 12 studies, while the bibliometric analysis and statistical distributions are based on the extended corpus of 71 articles, as well as the identification in the literature of auxiliary studies, considered by the authors of this paper to be reference points for the field.

7. Discussions

This work is aimed at researchers and practitioners in the fields of civil engineering, the structural industry, and infrastructure monitoring, as well as those interested in the integration of advanced sensor technologies and AI engineers in structural health assessment. The work is also addressed to design engineers, infrastructure managers, and preventive maintenance specialists who wish to implement SHM solutions in buildings and bridges. The work is also aimed at students and doctoral candidates who are exploring modern methods of damage detection and degradation prognosis. Also included in the target audience are IoT system developers, technology companies, and regulatory authorities interested in the sustainability and safety of civil constructions [100].

The results obtained in this paper show that the field of AI-assisted SHM is in an expansion phase. The quantitative analysis initially identified 292 articles in WoS, of which only 71 explicitly integrated the concept of AI in combination with SHM sensors. The report shows that although SHM is a mature field, the integration of AI models remains a narrow niche, but one that promises to develop in the future with the help of researchers.

The temporal distribution of publications shows growing interest from the academic community, demonstrated by the six articles published in 2020, a number that increased to 24 by 2025. The trend indicates an interest in maturing AI algorithms in this context. The qualitative analysis of the selected studies showed a correlation between the type of sensor used and the SHM task addressed. Accelerometers are the most widely used for damage detection and classification, especially when vibration-based applications are employed. CNN-type models applied to data from accelerometers produced the highest accuracy, reaching 99.65% in the case of multi-class structural damage classification. The values show that DL-type models are suitable for automatically extracting features from complex signals.

Visual sensors are used specifically for crack detection and surface degradation assessment. The analyzed studies reported accuracy values ranging from 83% to 94%. These results are obtained based on image analysis aimed at identifying crack detection. It should be mentioned that this quality is directly dependent on the quality of the visual data and the lighting conditions used to obtain the image.

In the case of fiber-optic sensors, the performance of SHM systems for medium- and large-sized defects corresponds to an F1-score exceeding 99%. For early-stage defects, the F1-scores are very low. These results show a limitation in the spatial resolution and sensitivity of current systems. The limitation identified in the detection of early defects has multiple causes. Firstly, the spatial resolution of the FBG networks and the sensor density affect the ability to capture local micro-deformation variations. Secondly, the signal-to-noise ratio masks the very small changes associated with small-sized defects. Additionally, performance is affected by the architecture of the algorithm used, especially in the case of models trained on data predominantly associated with medium or large-sized defects. A future research direction for researchers in the field is the use of hybrid sensor configurations (for example, combining FBG with PZT or MEMS sensors). Another recommendation from the authors of this study is the integration of Physics-Informed Machine Learning architectures, which can constrain predictions through physical laws and amplify sensitivity to subtle variations. This issue represents a field for the future development of SHM systems oriented toward preventive maintenance. Consequently, the development of hybrid sensor architectures coupled with advanced data processing methods is necessary. Unsupervised auto-encoder models and anomaly detection algorithms have the ability to identify deviations from the normal behavior of structures. In the analyzed articles, the detection rate exceeds 90%. They are less effective in classifying the severity of damage. Therefore, there is an applied distinction in complex engineering decision-making.

Another aspect discussed in the research concerns the comparison between traditional and AI-based models. Classical models use mechanical validation, but they are computationally expensive and difficult to adapt for complex structures. In contrast, AI models have high processing speed and automatic generalization capabilities, but they suffer from a lack of interpretability and dependence on large volumes of data. Integrating them into physics-informed ML and digital twin frameworks should represent a future direction of development for researchers who will continue to advance the field of SHM. The differences between traditional approaches and those based on AI are analyzed technically in relation to several parameters. Methods based on physical models require computational costs that cannot be ignored during the modeling and calibration phase. DL models are dependent

on large volumes of data and extremely costly computational resources, which are used in the training phase. The scalability of AI systems is superior in the case of extensive infrastructure, but the issues of quantifying uncertainty and explainability remain open across all domains where AI tools are used, being a highly debated issue at the moment. In the field of SHM, which is safety-critical, aspects such as model performance, the transition from experimental to practical use, explainable AI, uncertainty estimation for predictions, and the assessment of misclassification risk become essential elements in integration. The integration of interpretability mechanisms and uncertainty quantification methods should be a concern for the academic environment in the near future, as this represents a necessary direction for increasing trust in AI systems applied to civil infrastructures.

From an operational perspective, integrated sensor networks require design decisions that are not limited to model selection. Sensor placement, network topology, synchronization, sampling frequency, wireless communication, edge/cloud processing, latency, missing data, sensor faults, calibration, and long-term maintenance determine whether the AI output can support engineering decisions. Data fusion is particularly important because accelerometers, strain sensors, PZT sensors, FBG networks, cameras, LiDAR, and environmental sensors provide complementary information about dynamic response, deformation, surface damage, and operational conditions. Robust SHM deployment, therefore, depends on the joint configuration of the sensing network, data acquisition pipeline, AI model, and decision-support threshold.

The practical interpretation of AI outputs must also remain connected to structural mechanics. A predicted damage class, crack mask, anomaly score, or deformation forecast is useful only if it can be translated into reliability assessment, maintenance prioritization, uncertainty propagation, or an intervention decision. False alarms may generate unnecessary inspections and costs, while missed detections may allow degradation to progress. For this reason, AI-assisted SHM should be treated as a decision-support component that complements engineering judgment, physical modeling, and safety regulations rather than replacing them.

The paper presents a series of limitations determined by the selection of the literature, the analyzed time interval, and the open-access restriction. The analysis focuses on civil buildings and infrastructure, which means it does not cover industrial or special construction. The evaluation of AI algorithm performance is based on data reported by public studies, and the heterogeneity of datasets, sensors, structural systems, validation protocols, and metrics limits direct statistical normalization. Consequently, the review provides a structured qualitative synthesis rather than a meta-analysis. The restriction to open-access papers may exclude influential subscription-based studies and can affect the representativeness of the conclusions.

Regarding future research directions, the analysis indicates several potential future research directions, among which the development of hybrid sensors with enhanced resolution for detecting incipient defects stands out. Also noteworthy is the integration of AI algorithms with physical models to improve accuracy and expand SHM applications to residential buildings and historical infrastructure. The authors wish to explore historical buildings from an SHM perspective in future research, using IoT infrastructure. Additionally, the authors would like to explore the area of Digital Twins for degradation forecasting, especially for historical buildings. Implementing real-time monitoring systems with edge processing or adopting common standards for sensor interoperability and result reproducibility across different platforms and environments could represent a future research direction for researchers interested in this field.

Alongside all other fields that integrate AI-specific technologies, the implementation of AI-based SHM systems raises the same ethical and regulatory issues. In a safety-critical

field, an erroneous classification can lead either to unnecessary interventions or to the failure to identify a dangerous degradation. Therefore, regulated standards for model validation, experiment reproducibility, and traceability of automated decisions are necessary. Quantifying the uncertainty of predictions and integrating explainable AI mechanisms are intensively debated at both European and international levels to increase trust in automated systems. Legal responsibility in the case of AI-assisted decisions represents an emerging issue that must be integrated into future regulatory frameworks for civil infrastructure.

8. Conclusions

The conclusions drawn in this paper concern the possibility of monitoring SHM using AI technologies through sensors. The Conclusions Section synthesizes the main findings of the analysis, without reiterating the interpretative details or extensive discussions presented earlier. In this section, the emphasis is placed on the results and the central contributions of the review.

A systematic analysis of the literature from 1 January 2020 to 31 December 2025 shows an increasing interest in these solutions from the academic community. The results show that the performance of these systems is correlated with the type of sensor used and the monitoring task.

Several accelerometer-based and vision-based studies report high performance for damage detection and classification, including values above 95% under controlled, benchmark, simulated, or study-specific validation conditions. These results should not be interpreted as universal performance guarantees for operational structures. Fiber-optic sensors enable deformation and strain monitoring, but the detection of very small incipient defects remains problematic and is affected by sensor density, spatial resolution, noise, and model architecture.

Comparing it with traditional methods highlights a paradigm shift that addresses non-linearities and large data volumes. Neither paradigm is sufficient on its own, which is why hybrid approaches like physics-informed ML and Digital Twins are shaping future research directions.

Integrating high-performance sensors into AI algorithms is a strategic direction in SHM. Future research should focus on detecting early defects, increasing the interpretability of AI models, obtaining suitable data for these models, managing infrastructure that can utilize AI models, and expanding applications to residential buildings. These directions stem from the fact that the aforementioned topics are currently insufficiently explored.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/buildings16122299/s1>, PRISMA_2020_checklist.

Author Contributions: Conceptualization, C.-M.R.; methodology, C.-M.R., A.S. and C.P.; software, C.-M.R. and A.S.; validation, C.-M.R., A.S. and C.P.; formal analysis, C.-M.R., A.S. and C.P.; investigation, C.-M.R., A.S. and C.P.; resources, C.-M.R., A.S. and C.P.; data curation, C.-M.R. and A.S.; writing—original draft preparation, C.-M.R., A.S. and C.P.; writing—review and editing, C.-M.R., A.S. and C.P.; visualization C.-M.R., A.S. and C.P.; supervision, C.-M.R. All authors have read and agreed to the published version of the manuscript.

Funding: This research was partially funded by the Petroleum-Gas University of Ploiesti.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The review process generated search strings, database counts, screened records, extracted variables, classification criteria, and inclusion/exclusion decisions.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

1D	One-dimensional
3D	Three-dimensional
AE	Acoustic emission
AI	Artificial intelligence
ANN	Artificial Neural Network
AUC	Area Under the Curve
BOTDA	Brillouin Optical Time-Domain Analysis
CNN	Convolutional Neural Network
DAS	Distributed Acoustic Sensing
DIC	Digital image correlation
DL	Deep Learning
DNN	Deep Neural Network
EMA	Experimental Modal Analysis
EMI	Electromagnetic interference
FBG	Fiber Bragg grating
GRU	Gated Recurrent
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
IR	Infrared
LiDAR	Light Detection and Ranging
LVDT	Linear Variable Differential Transformer
MAE	Mean Absolute Error
mAP	Mean Average Precision
MAPE	Mean Absolute Percentage Error
MEMS	Micro-electro-mechanical systems
ML	Machine Learning
NN	Neural Network
OFS	Optical Fiber Sensor
PIML	Physics-Informed Machine Learning
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PZT	Piezoelectric transducer
R ²	Coefficient of determination
RF	Random Forest
RGB	Red, green, blue
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic
RRMSE	Relative Root Mean Square Error
SHM	Structural health monitoring
SVM	Support Vector Machine
TF	Transmissibility function
TTF	Temperature–time factor
UAV	Unmanned Aerial Vehicle
VMD	Variational Mode Decomposition
WoS	Web of Science
WSN	Wireless Sensor Network
YOLO	You Only Look Once

Appendix A

Table A1. The 12-study methodological core.

SHM Task	AI Model	Reference
Crack image classification and object detection	DL	Shao et al. [69]
Time-series regression	DL	Liu [71]
Vibration-based structural health monitoring	NN	Wijethunga et al. [75]
Multi-class damage classification (5–15 classes)	DL	Dabbous et al. [90]
Binary damage classification (retrofitting state)	ML	Presno Vélez et al. [91]
Structural response reconstruction	NN	Wang et al. [92]
3D displacement measurement	DL	Shao et al. [93]
Damage detection & localization (small to large defects)	ML	Herrera et al. [95]
Damage localization and size classification	ML	Perfetto et al. [96]
Anomaly detection and severity assessment	DL	Liu et al. [97]
Damage detection & localization	ML	Gunes and Gunes [98]
Binary anomaly detection	ML	de Alteriis et al. [99]

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